

Chase Mateusiak

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Skills

Fluent in R and Python with years of Linux/HPC experience (SLURM). Nextflow user since 2019, including a published nf-core pipeline. Deployed and administer Seqera Platform on AWS for two labs. Experienced with containerization, Git/GitHub, and cloud infrastructure. Build and maintain automated data processing pipelines across multiple omics data types.

Work

Bioinformatics Scientist

Washington University, St. Louis, *2020–2026*

Primary responsibilities

- Implement established, recently published, and novel analysis algorithms on bulk RNA-seq, various transcription factor binding modalities (Calling Cards, ChEC-seq, ChIP-seq) and sequence analysis in both single data type and multi-omic modes
- Responsible for data processing and QC of sequencing experiment data for multiple labs, including a large \$80 million multi-site collaboration
- Provide computational support to collaborators across organisms (yeast, mouse, human)

Notable achievements

- Developed a novel method of inferring transcription factor interaction at promoters, integrating both transcription factor binding and overexpression and knockout (Mateusiak et al, currently out for review 2026)
- Collected, harmonized, databased all 11 of the major yeast transcription factor binding and perturbation response datasets (Mateusiak et al, expected publication 2026). Notably, the data is hosted using a data lake architecture on huggingface, a machine learning platform that provides free hosting. I developed both a programmatic library and shiny app that provides researchers easy access to this rich data
- Developed, published and released on nf-core a Nextflow workflow for processing Calling Cards data in both yeast and mammals
- Implemented and manage the cloud architecture for two labs, including providing access to Seqera Platform
- Developed independent collaborations resulting in two publications
- Provided computational support to a clinical group by implementing and evaluating the performance of four of the leading circadian rhythm inference algorithms
- Developed the Python curriculum implementation for Algorithms for Computational Biology, a 500-level course in the Washington University CS department
- Trained and directly supervised 5 undergraduate and master's students in machine learning based methods of multi-omics data analysis

Research Assistant

University of California, San Francisco, *2019*

- Processed RNA-seq data from human skin cells and tick
- Compiled transcriptome annotations for *Ixodes scapularis*

- Processed data from a deep mutational scan screen in *E. coli*

Teacher

San Francisco Unified School District, 2012–2019

- Responsible for the academic progress of 115 students per year in English and math

Civil Affairs Sergeant

US Army Reserve, 2011–2019

- Trained in developing local relationships, identifying achievable development projects, and coordinating allied forces activities with local government and non-profit organizations

Teacher

US Peace Corps, Turkmenistan, 2008–2010

- Developed and implemented an interactive English as a Foreign Language curriculum that improved speaking and vocabulary skills across eight classrooms for over 250 students

Education

- M.S. Computer Science — Washington University, St. Louis, 2022
- B.A. English Literature and Language — University of Michigan, Ann Arbor, 2008

Publications

Mateusiak, C., Jia, E., Plaggenberg, J. N., Erdenebaatar, Z., Wang, Y., Shively, C., Liao, G., Mitra, R. D., & Brent, M. R. (2026). Ubiquitous functional synergy partially explains why most transcription factor binding is non-functional. *bioRxiv* (preprint). <https://doi.org/10.64898/2026.01.19.700460>

Karagiannis, T. T., Mateusiak, C., Acharya, S., Jung, W. J., Liao, S., et al. (2026). Whole blood transcriptional signatures of age and survival identified in long life family and integrative longevity omics studies. *GeroScience*. <https://doi.org/10.1007/s11357-025-02090-x>

Abid, D., Brown, H., Mateusiak, C., Doering, T. L., & Brent, M. R. (2025). Mapping the transcriptional regulatory network of a fungal pathogen by exploiting transcription factor perturbation. *mBio*. <https://doi.org/10.1128/mbio.02797-25>

Kang, Y. S., Jung, J., Brown, H. L., Mateusiak, C., Doering, T. L., & Brent, M. R. (2025). Leveraging a new data resource to define the response of *Cryptococcus neoformans* to environmental signals. *Genetics*, 229(1), 1–29. <https://doi.org/10.1093/genetics/iyae178>

Yen, A., Mateusiak, C., Sarafinova, S., Gachechiladze, M. A., Guo, J., Chen, X., Moudgil, A., Cammack, A. J., Hoisington-Lopez, J., Crosby, M., Brent, M. R., Mitra, R. D., & Dougherty, J. D. (2023). Calling Cards: a customizable platform to longitudinally record protein-DNA interactions over time in cells and tissues. *Current Protocols*, 3(9), e883. <https://doi.org/10.1002/cpz1.883>

Ring, K., Couper, L. I., Sapiro, A. L., Yarza, F., Yang, X. F., Clay, K., Mateusiak, C., Chou, S., & Swei, A. (2022). Host blood meal identity modifies vector gene expression and competency. *Molecular Ecology*, 31(9), 2698–2711. <https://doi.org/10.1111/mec.16413>

Radkov, A., Sapiro, A. L., Flores, S., Henderson, C., Saunders, H., Kim, R., Massa, S., Thompson, S., Mateusiak, C., Biboy, J., Zhao, Z., Starita, L. M., Hatleberg, W. L., Vollmer, W., Russell, A. B., Simorre, J. P., Anthony-Cahill, S., Brzovic, P., Hayes, B., & Chou, S. (2022). Antibacterial potency of type VI amidase effector toxins is dependent on substrate topology and cellular context. *eLife*, 11, e79796. <https://doi.org/10.7554/eLife.79796>